Lab 2 Network Utility Programs and DNS Protocol

This lab is an **individual** lab and can be done at your own time and place. it is important to start the lab before the lab session, since time of 2h might not be enough to complete the lab. During the lab session, you can get instant support/help from the instructor. If you have completed the this lab and do not have questions, it is not required to be present in this lab session.

1. Intended Learning Outcomes

At the end of this lab activity, you should be able to

- Use common network utility (**ipconfig, nslookup**) programs to find network configuration parameters
- Use **Wireshark** to capture and analyse DNS messages
- Explain how DNS **name resolution** works and its applications.

2. Tasks

There are 4 tasks in this lab. Follow the instructions, record the data, and answer the questions.

We start with some background information about protocols and data encapsulation.

When upper layer protocols communicate with each other, data at the sending host flows down the TCP/IP protocol layers and is encapsulated into a protocol data unit at a lower layer, and finally encapsulated in a Layer 2 frame. For example, DNS message is often transported by UDP protocol on layer 4. So a DNS message at the sending host is encapsulated in a UDP protocol data unit(PDU); the UDP PDU is then encapsulated in an IP packet, and the IP packet is encapsulated finally in a layer 2 frame. The frame composition is dependent on the media access type, or the network. For example, if the media access is Ethernet, then the Layer 2 frame encapsulation will be Ethernet. When learning about data encapsulation and protocol operations, it is helpful to analyze the header information found in the protocol data units. The DNS protocol operation, network utility programs like **ipconfig** command will be examined in this lab.

There are **conventional names for the protocol data units** for different layer protocols. Answer the following question:

Question 1. What are names for the protocol data units (PDUs) for layer 4, layer 3, and layer 2 protocols in TCP/IP reference model by filling the follow form: